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Surgery for hilar cholangiocarcinoma: The Leeds experience

E. Hidalgo [a], S. Asthana [a], H. Nishio [a], J. Wyatt [b], G.J. Toogood [a], K.R. Prasad [a], J.P.A. Lodge [a,*]

[a] *Hepatobiliary and Transplant Unit, St James's University Hospital, Leeds LS9 7TF, UK*
[b] *Department of Pathology, St James's University Hospital, Leeds LS9 7TF, UK*

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Abstract

Aim: To review the experience with hilar cholangiocarcinoma and to determine the results of a radical surgical approach in a UK centre.
Methods: A 10-year review of all patients treated surgically for proximal bile duct carcinoma at a single surgical unit was conducted. Patient demographics, disease details and histopathology reports were reviewed. From January 1993 through December 2003, 106 patients were admitted with the diagnosis of hilar cholangiocarcinoma and 61 patients received surgical exploration.
Results: Tumours were staged as follows (UICC 6th edition): stage IB, $n = 10$ IIA, $n = 9$; IIB, $n = 20$; III, $n = 8$; IV, $n = 14$. Out of 61 patients, 44 had a resection (3 bile duct resection alone, 41 liver resection with bile duct resection), 5 were considered unresectable and 12 underwent liver transplantation (LT). The caudate lobe was excised in 34 of the patients and regional lymphadenectomy was systematically carried out. Para-aortic lymphadenectomy was performed in 17 cases. Portal vein resection was needed in 17 and hepatic artery resection was performed in 4 cases. Negative histologic margins (R0) were achieved in 20 patients and microscopic margin involvement (R1) was seen in 16. In the remaining 8 resected patients, localised metastasis were found (peritoneal deposits in 2, liver metastasis in 4 and positive para-aortic lymph nodes in 2); nevertheless the resection was performed and it was considered R2. Overall survival at 3 and 5 years for patients who underwent a resection was 43% and 28% including postoperative deaths. The 1-, 3- and 5-year actuarial survival rates for patients who underwent R0 resection were 78%, 64% and 45% respectively, including the postoperative deaths ($n = 3$). The median survival time was 41.1 months. The 1-, 3- and 5-year actuarial survival rates for R1 resection and R2 were 60%, 26%, 26% and 25% and 0% respectively, while the median survival time for these groups was 15.4 and 6.8 months respectively. The actuarial survival rate at 1, 3 and 5 years for well-differentiated tumours (G1) was 73%, 54% and 40% (median 39.7 months). The figures for G2 were 60%, 48% and 0%. The figures for G3 (poorly differentiated) were 16% and 0% at three years ($p = 0.03$).The overall survival at 3 and 5 years for those patients who had a liver transplant was 41% and 20% including early postoperative mortality. The tumour grading (presence of poorly differentiated tumour) was found to be the only independent factor affecting the survival time producing a hazard ratio of 4.3 ($p = 0.0034$, 95% confidence interval 0.1007–6.342).
Conclusions: Radical surgical resection is the best treatment for hilar cholangiocarcinoma. R0 resection provides acceptable 5-year survival, but R1 resection may also provide acceptable palliation. In our experience TNM stage and tumour grade were the main determinants of long-term survival.
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Keywords: Hilar cholangiocarcinoma; Surgery; Radical resection

Introduction

Biliary malignancies constitute about 2% of all cancers found at autopsy.[1] Perihilar or Klatskin tumours represent about 50–80% of these cases.[2–6] The prevalence of biliary cancer (gallbladder and bile ducts) in the UK is 2.8 cases per 100,000 females and 2 cases per 100,000 males, while cholangiocarcinoma (CC) causes an estimated 1000 deaths annually.[7]

The surgical management of such tumours is extremely challenging because of the anatomic location (proximity of the vasculature). However, aggressive surgical resection with extended hepatectomy, lymphadenectomy and vascular reconstruction may offer the only possible chance of cure, as adjuvant therapeutic modalities play a limited role in its management.[8–11] Our centre has practised

* Corresponding author. Tel.: +44 113 206 4890; fax: +44 113 244 8182.
E-mail address: peter.lodge@leedsth.nhs.uk (J.P.A. Lodge).

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a policy of aggressive surgical resection in patients with advanced locoregional disease in the absence of distant metastases over the past 10 years. This study was conducted to analyse the early and long-term results of this approach in Leeds.

Methods

Demographics

A review of all patients who underwent a surgical exploration for proximal bile duct carcinoma was carried out. Patients with an adenocarcinoma arising at the confluence (hilar) or involving it (perihilar) were included; gallbladder cancers were not included in this cohort. From January 1993 through December 2003, 106 patients were admitted at this institution with the diagnosis of proximal bile duct cancer. Sixty-one patients with cholangiocarcinoma received surgical exploration. The median follow-up was 21.7 months. The mean age was 53 years (ranging from 27 to 81); 60% of the patients were males and 9/61 had an underlying liver disease (6 primary sclerosing cholangitis without cirrhosis). On 6 occasions, an attempted liver resection was performed at another institution. All of them were considered resectable and underwent a further operation.

Patient evaluation

Forty-four out of 61 patients were referred from other centres. Most of these patients had undergone an invasive cholangiogram (mainly ERCP) and subsequent stenting. In order to assess tumour extension a contrast-enhanced magnetic resonance imaging scan was obtained. If jaundice was present a biliary drainage was performed. The commonest method at this centre was ERCP and retrograde stenting. One-third of these patients also required PTC for further drainage. Six patients underwent a combined procedure when the initial drainage was considered insufficient (cholangitis or persistently high bilirubin). A chest/abdomen CT was systematically obtained.

None of the reviewed patients underwent a preoperative portal vein embolisation. Since 2000, laparoscopic staging has been used to assess peritoneal extension of disease. Six patients had a negative laparoscopy performed elsewhere. An exploratory laparotomy was performed in 7 out of 12 cases considered for a liver transplant.

Criteria for resectability

The disease was considered unresectable in the presence of demonstrable peritoneal disease, metastatic liver involvement outside the field of resection, or macroscopic para-aortic nodal disease. Vascular invasion was not considered as a contraindication.

Histology

All the specimens were reviewed by the same pathologist. Frozen section assessment of surgical margins was carried out if there was a clinical suspicion of margin involvement.

Statistical analysis

Data is expressed as percentage or mean ± standard error of the mean. Median values are also given. Percentages were compared by the chi-square test, and means were analysed by Fisher's test or the Mann–Whitney test. Survival curves were obtained by the Kaplan–Meier (log-rank) test. Cox's proportional hazard survival analysis was used to determine which parameters affected survival. A p value of <0.05 was considered significant. Statistical analysis was carried out using SPSS software Version 11.0 (SPSS Inc, Chicago, IL).

Results

Surgical procedures

Sixty-one patients were included in this study. Forty-four patients underwent a resection and 12 patients had a liver transplant (LT) (Table 1). Five were considered unresectable at the time of the laparotomy (extensive peritoneal disease in 4, multiple liver metastases in 1) and underwent palliative radiological stenting. These patients did not have a laparoscopy (early part of study). Resectability rate was finally 53% (56/106).

Of those patients who underwent a potentially curative resection, 17 required portal vein resection and 4 had a hepatic artery resection, while 3 required a simultaneous vein and arterial reconstruction. All patients had a regional lymphadenectomy (hepatoduodenal, retroduodenal and celiac

Table 1
Type of surgical procedure

	Patients	PV resection	HA resection
Laparotomy and biopsy	5		
Liver transplant	12		
Resection	44		
CBD excision	3	1	2
Liver resection	41		
plus S1 excision	1		
Left hemihepatectomy + S1	7	3	
Left trisectionectomy + S1	12	6	
Right hemihepatectomy	4		
Right trisectionectomy		7	2
+S1 excision	14		
No S1 excision	2		
Whipple's + S1	1		
Total	61	17	4[a]

PV, portal vein; HA, hepatic artery.
[a] Combined PV and HA resection in 3 patients.

axis lymph node stations); in 17 patients lymphadenectomy was extended to the para-aortic chain with clearance of nodes down to the aortic bifurcation (early part of the study) or the inferior mesenteric artery (more recently). Particular attention was paid to neurectomy to the right of the superior mesenteric artery, around the celiac axis, the hepatic artery and its branches. Segment 1 was removed in all left hemihepatectomies and almost all right trisectionectomies.

LT was indicated when the tumour was locally advanced with no evidence of extrahepatic disease and unlikely to be completely excised with resectional surgery. The majority (7 patients) had a pre-transplant exploratory laparotomy.

Perioperative complications after resection

The mean hospital stay was 20 days (median was 17 days) (Table 2). Twenty-six of the patients undergoing resection experienced some form of postoperative morbidity (59%). The morbidity for LT patients was not included in this analysis.

Three of 44 patients undergoing resection died at the hospital within the first 30 days (perioperative mortality 6.8%). All of them had undergone a left hepatic trisectionectomy and none of them had liver insufficiency as a primary cause of death, although secondary liver failure developed following the initial event. The mortality rate at 3 months was 11.3%.

Table 2
Perioperative complications

Type of complication	No. of events
Pleural effusion[a]	2
Wound infection	4
Wound dehiscence	2
Respiratory infection	2
Upper GI bleeding	2
Acute renal failure	4
Portal vein thrombosis	2[d]
Hepatic artery thrombosis	1[d]
Right hepatic vein thrombosis	1
Pulmonary embolism	1
Myocardial infarction	1
Intra-abdominal abscess[b]	6
Intra-abdominal haemorrhage	4
Small bowel fistula	1
Bile leak	1
Acute pancreatitis	1
Liver failure[c]	8
MRSA sepsis	7
Reoperation	6
Haemorrhage	3
Wound dehiscence	2
Bile leak	–
SB fistula	1
Postoperative mortality (<30)	3

GI, gastrointestinal; MRSA, methicillin-resistant *Staphylococcus aureus*.
[a] Pleural effusion that required a chest drain.
[b] Intra-abdominal collection that required a drainage.
[c] Jaundice and encephalopathy.
[d] Only one patient had vascular reconstruction.

Histopathology

All the patients initially included (61) were classified as follows: stage IB 10, IIA 9, IIB 20, III 8, and stage IV 14 (UICC 6th edition) (Table 3). Lymph node metastases were present in 58% patients and vascular involvement was present in 33% of the cases. Among the 44 patients who had a resection, R0 resection was achieved in 45%; 36% of the patients had a R1 resection and 18% were reported as a R2 resection. In this last subset, 8 patients were found to have limited metastatic disease at the time of the laparotomy (liver metastasis in the resected liver 4, peritoneal deposits 2, and positive para-aortic lymph nodes 2). These conditions were unknown preoperatively. However, a major operation was carried out mainly because of relatively young patient age (mean 51 years). On analysis of transplanted group of patients 75% had an R0 resection.

Table 3
Comparison between resected and transplanted patients

Parameter	Transplanted ($n = 12$)	Resected ($n = 44$)	p value
Gender			n.s.
Male	8	24	
Female	4	20	
Age (years)	$47.7 \pm 8{,}4$	54.7 ± 11	0.012
Tumour size (mm)	36.7 ± 20	30.03 ± 16	n.s.
Liver disease			n.s.
PSC	4	3	
SBC	1		
UICC 6th ed. Stage			n.s.
I–II	10	29	
III–IV	2	15	
T factor			n.s.
T1–2	5	15	
T3–4	7	29	
N factor			n.s.
N0	5	20	
N1	7	24	
Presence of metastasis			n.s.
No	11	36	
Yes	1	8	
Residual tumour status			0.051
R0	9	20	
R1–R2	3	24	
Tumour grading			n.s.
G1 (Well diff.)	8	24	
G2 (Mod. diff.)	2	11	
G3 (Poorly diff.)	2	9	
Perineural invasion			NA
No	–	4	
Yes	12	38	
Vascular Invasion			n.s.
No	9	27	
Yes	3	17	

n.s.: not significant; NA: not available.

Survival analysis

Since no patient with recurrence survived, the analysis refers only to patient survival.

Overall survival rates

The overall 1-, 3- and 5-year survival rates for all the patients entered in this study were 58%, 35% and 22%, with a median survival of 16.1 months(Fig. 1). The overall survival rate for the subset of 50 patients who underwent palliative treatment (no surgery $n = 45$ or failed surgical exploration $n = 5$) was 13% at 1 year, with a median survival of 4.13 months.

The overall 1-, 3- and 5-year survival rate for the resected and the transplanted group of patients were 60%, 43%, 28% and 58%, 41% and 20%, respectively, without significant differences. The further analysis is limited to the 44 patients who underwent a resection. Table 4 shows the univariate analysis regarding the main clinicopathological parameters.

Survival outcome according to resection margins

The 1-, 3- and 5-year actuarial survival rates for patients who underwent R0 resection were 78%, 64% and 45% respectively, including patients that died perioperatively (median survival time 41.1 months) (Fig. 2).

The 1-, 3- and 5-year actuarial survival rates for R1 resection and R2 were 60%, 26%, 26% and 25 and 0% respectively (median survival time 15.4 and 6.8 months respectively). The overall log-rank test significance was 0.046 although R0 survival did not differ statistically from R1 resection.

Survival outcome according to lymph node metastases

In the resection group, the 1-, 3- and 5-year actuarial survival rates for patients who had histologically negative lymph nodes were 65%, 57% and 38% (median survival time 36.1 months). The survival figures with positive lymph nodes were 57%, 33% and 22% respectively (median survival of 15.4 months, $p =$ not significant). However, when the presence of nodal disease was analysed including the 12 LT patients, the differences achieved significance (median survival 39.7 vs. 12.4 months, $p = 0.05$).

Survival outcome according the perineural invasion

This did not influence long-term survival in this study.

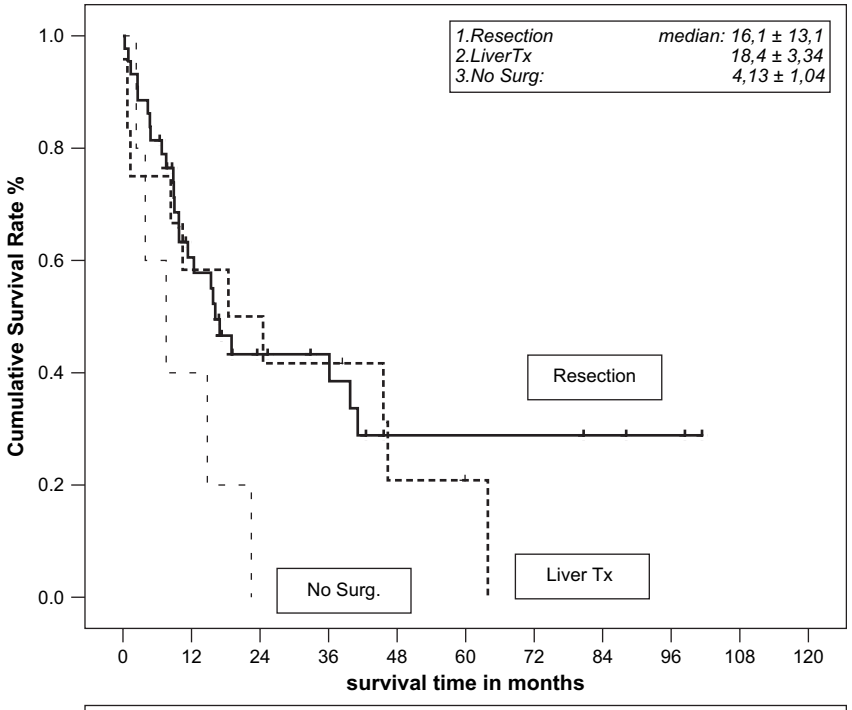


Figure 1. Overall survival rates according to the procedure.

Table 4
Analysis of the clinicopathological parameters

Parameter	*n*	Event	Mean	Median	*p* value
Type of procedure					n.s.
Hemihepatectomy	12	6	36.04	19	
Trisectionectomy	29	16	41.8	15.4	
CBD resection	3	2	35.65	12.4	
UICC Stage					0.07
IB	9	3	64.3	19.5	
IIA	9	3	56.9	11.3	
IIB	14	7	25.4	16.1	
III	7	5	12.6	12.5	
IV	8	7	14.3	6.8	
UICC Stage					0.013
I–II	29	13	50.13	36.1	
III–IV	15	12	15.5	9.3	
T factor					n.s.
T1	1	0	NA	NA	
T2	15	7	50.9	36.1	
T3	20	12	33.4	9.8	
T4	8)	6	19.5	9.7	
Residual tumour status					0.046
R0	20	8	55.7	41.1	
R1	16	10	34.6	15.4	
R2	8	7	14.3	6.8	
Nodal disease					n.s.
N0	20	9	49.8	36.1	
N1	24	16	20.6	15.4	
Presence of metastasis					0.029
No	36	18	46.1	19	
Yes	8	7	14.3	6.8	
Tumour grading					0.03
G1 (well diff.)	24	12	51.8	39.7	
G2 (moderately diff.)	11	6	21.2	12.4	
G3 (poorly diff.)	9	7	7.6	6.8	
Perineural invasion					n.s.
No	6	2	40.4	9	
Yes	38	21	41.4	16.4	
Vascular invasion					0.03
No	27	12	48.2	36.1	
Yes	17	13	23.7	8.9	

Data are reported in months; event = patient's death; *p* value calculated with log-rank test. Perioperative deaths are included. UICC refers to the 6th edition.

Survival outcome according the T category (pTNM, UICC 6th edition)

The 1.5-year survival in the T1 subgroup was 100% and 100%; for the T2 subgroup the survival rates at 1, 3 and 5 years were 86%, 60% and 36% (median survival 36.1 months). Patients who had a T3 tumour had 42%, 28% and 28% survival rates at 1, 3 and 5 years. Finally the T4 group had a survival rate of 50% and 0% at 1 and 3 years (log-rank not significant).

Survival outcome according the tumour grading

The actuarial survival rate at 1, 3 and 5 years for well-differentiated tumours (G1) was 73%, 54% and 40% (median 39.7 months). The figures for G2 were 60%, 48% and 0%. The figures for G3 (poorly differentiated) were 16% and 0% at 1 and 3 years ($p = 0.03$).

Survival outcome according the presence of metastasis

The overall survival in the absence of distant metastasis was 69%, 47% and 35% at 1, 3 and 5 years. A total of 8 patients presented with metastasis and underwent a resection. In this subset, the survival rates were 25% and 0% at 1 and 3 years ($p = 0.029$).

Survival outcome according the pTNM classification (UICC 6th edition)

Survival in patients with stage IB disease at 3 and 5 years was 53% and 53%. For stage IIA, the survival rate at 3 and 5 years was 50% and 50%; in patients with stage IIB survival was 55% and 34% at 3 and 5 years respectively. For tumours with stages III and IV, the survival time at 1 and 3 years was 42% and 0%, and 25% and 0%, respectively ($p = 0.07$). The differences were more accentuated when stages I–II disease were compared vs. stages III–IV disease, with a median survival of 36.1 vs. 9 months respectively ($p = 0.013$).

Survival outcome according the presence of vascular invasion

When histological invasion of portal vein was present survival rates at 1, 3 and 5 years were 41%, 28% and 18%: median survival time 8.9 months, vs. 36.1 months in patients without portal vein invasion ($p = 0.03$). Portal vein resection and reconstruction did not have a statistically worse outcome in the absence of histologic involvement.

Survival outcome according the pattern of residual tumour

The different types of R1 resection were classified as either hepatic duct margin positive (7) or transection plane positive (9). Four patients had both hepatic duct and parenchymal margin involvement. The survival rates did not differ amongst them.

Multivariate analysis

A multivariate analysis was conducted on variables found to achieve significance on univariate analysis (Cox's regression model, Backward stepwise, likelihood ratio; contrast Helmert). The tumour grading (poorly differentiated tumour) was found to be the only independent factor affecting the survival time producing a hazard ratio of 4.3 ($p = 0.0034$, 95% confidence interval 0.1007–0.6342).

Discussion

This study represents the largest published CC series from the UK. Most large series with long periods of follow-up are from Japanese centres and other eastern countries.[12–22] Surgical management of these tumours has tended to follow two distinct pathways: surgical resection or transplantation.[23–30]

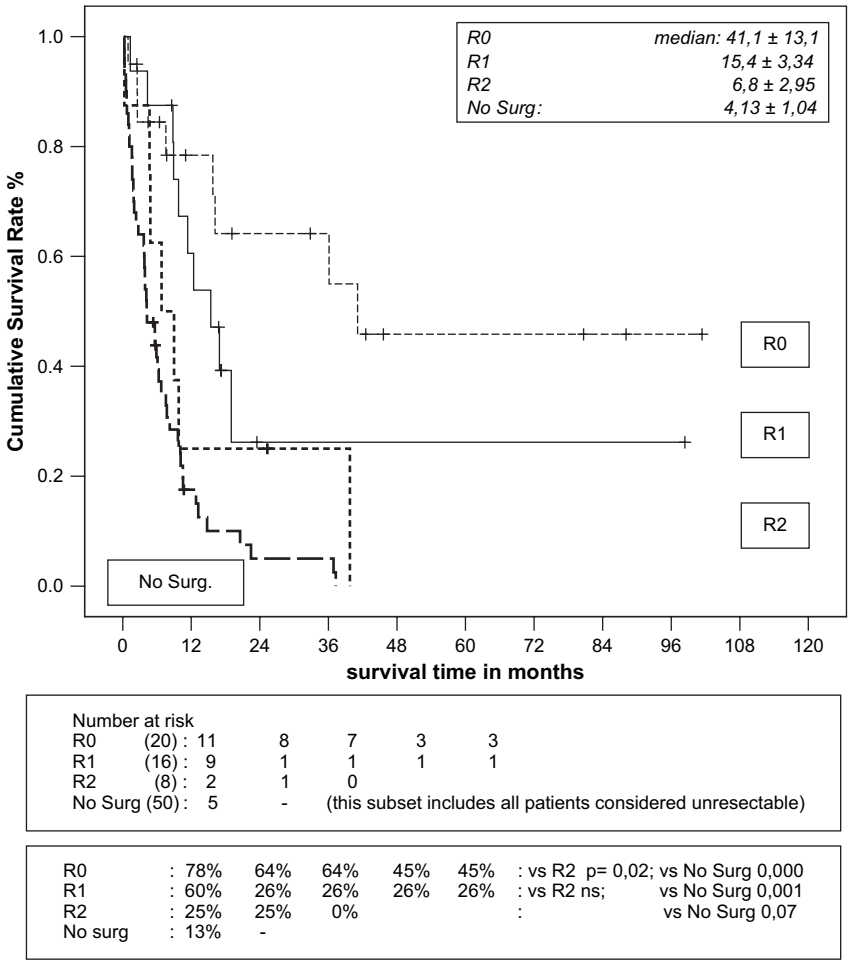


Figure 2. Overall survival rates according the residual tumour pattern.

Liver transplantation

This remains controversial because of poor long-term survival.[31–34] All liver transplants for CC at this centre were performed between 1991 and 2001, as after 2001 LT for CC was discontinued. The age of the recipients was significantly lower among the transplanted patients. The tumour size was bigger among these patients although it did not reach statistical significance. R0 resection rate was significantly higher among the transplanted patients, yet these patients fared worse than with resection indicating that transplantation alone is unlikely to be beneficial. The additional requirement of immunosuppression in these patients may worsen the outcomes in patients with unfavourable biological factors.[33–35] Recent data from the Mayo Clinic suggest a role for neoadjuvant chemoradiation in this patient population.[36]

Liver resection

We believe resection and bile duct excision is the procedure of choice.[15–18,20–23,32,37,38] Biliary tree excision alone was only performed when the patient was considered unfit. At laparotomy, 5 patients were considered unresectable because of extensive metastatic disease, and 8 underwent a palliative resection despite localised metastatic disease. Therefore only 36 patients had a potentially curative resection.

In our centre the main criteria precluding resection was the presence of distant peritoneal disease or liver metastasis outside the field of resection. Of all the patients who underwent resection, 41 had a liver resection and 3 had a biliary tree excision alone.

Management of patients undergoing resection

None of our patients underwent preoperative portal venous embolisation (PVE). PVE has been advocated as a means of encouraging growth of the liver remnant and a few major centres perform a routine PTC followed by PVE when a major hepatectomy is contemplated.[22,23,25] However, the incidence of liver failure quoted by these authors after PVE did not differ grossly from figures in this study. The high incidence of PV thrombosis may explain this similar outcome. The incidence of transient liver insufficiency was 18%, defined as an increased serum bilirubin

level accompanied by encephalopathy after the procedure. The overall morbidity reported in literature ranges between 20% and 65%, varying widely depending upon the extent of liver resection performed. The overall perioperative mortality within the first 30 days in this group was 6.8%. This compares favourably with reported mortality rates in literature, which ranges from 0% to 25%.[6,20,21,41–44] The lowest rate has been recently reported by Seyama et al.[22]

Caudate lobe was removed[38–40] in 77% of all liver resections and vascular reconstruction performed in 19 patients. All patients underwent a systematic celiac lymphadenectomy and an additional para-aortic lymphadenectomy was performed in 17 patients. This radical approach carries a high morbidity, being in our experience 59%.

Results of liver resection

Analysis of the histological results revealed a strikingly low rate of R0 resection. However, up to 67% of these patients would have been classified as stage IV according the former 5th edition,[45] with 23% of them being stage IVB.

On the other hand, 18% of resections were considered R2 because of localised metastatic deposits. These patients did not benefit from more aggressive surgery and would have been considered non operable and stented if the preoperative assessment had identified them accurately. Radical resection for Stage IV disease can probably only be justified in a highly selected group of patients, possibly those with para-aortic lymph node metastasis. Para-aortic nodal metastasis may be present in around 17% of patients and is associated with a poor prognosis. Whether systematic para-aortic lymphadenectomy affects the survival or not is not certain.[46,47] In our series only 2 patients were found to have para-aortic disease and therefore no conclusions can be drawn, although one of them is still alive 2 years after his operation.

The subset of patients who underwent R1 resection poses an interesting management problem, mainly because of the limits of aggressive surgical resection. Several tumours treated by resection had extensive biliary involvement and vascular encasement, which may be relative contraindication for aggressive resection.[20,23] Twelve patients had involvement of the hepatic duct margin, either alone or involving the hepatic parenchymal or vascular transection plane as well. Among the group with R1 resection, 10 of 16 underwent a right or a left trisectionectomy. Neuhaus et al. have suggested that right extended resections may offer a better choice of achieving R0 resection.[17] This was not borne out in this study, either in terms of the R0 rate or long-term survival. Extension of tumour along the hepatic duct determines the type of resection that needs to be done rather than performing a right trisectionectomy routinely. It is important to resect as far as technically is possible and ideally demonstrate that the remnant has no residual disease. Routine frozen section of the margins may be helpful, but interpretation remains challenging. In this study, the group of patients with R1 resection had a median survival time of 15.4 months, which was significantly better than the non-resected patients ($p = 0.001$).

Prognostic factors on univariate analysis in this study were the presence of residual tumour after resection, histological differentiation, vascular invasion and the presence of metastasis. On multivariate analysis, only the histological differentiation had a significant effect on the long-term survival: poorly differentiated tumours had a hazard ratio up to 4 times more than that for well-differentiated ones. The presence of vascular invasion did not reach statistical significance, most likely due to the small number of cases. However, larger series have shown that portal venous involvement influences survival.[23] Assessment of vascular involvement preoperatively is not very specific, and no patients should be excluded from a potential curative resection based on equivocal preoperative imaging alone, since up to 30% of them will have no histological invasion following the portal vein resection.

The results of the multivariate analysis in this cohort of patients are consistent with the existing literature.[13,17,21,23,40] and reflect the tumour's biological behaviour. Not only patients with residual disease but also those with poorly differentiated carcinomas may benefit from adjuvant treatment.[43]

In conclusion, radical surgical resection is the best treatment option for hilar cholangiocarcinoma. R0 resection provides acceptable five-year survival, although several other variables such as the histological tumour differentiation influence the prognosis. R1 resection may also provide acceptable palliation in our experience and should be considered if it can be performed with an acceptable low perioperative morbidity and mortality.

Conflict of interest

None of the authors participating in this study has any financial or personal relationship whatsoever with any kind of organisation or people that might have influenced or biased the content of this paper.

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